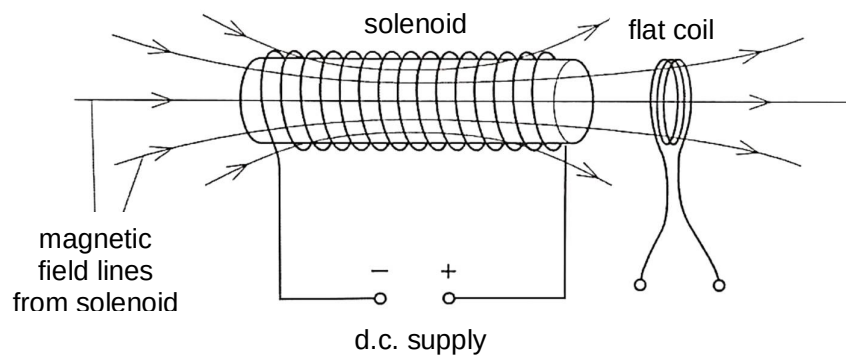
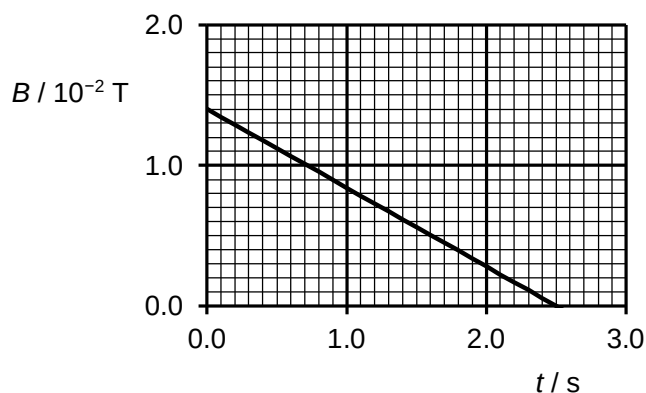


- 6 (a) A solenoid is connected to a variable d.c. supply as shown in Fig. 6.1.



**Fig. 6.1**

A flat coil of diameter 6.4 cm and 180 turns is placed close to one end of the solenoid. The coil has a resistance of  $1.5 \Omega$ . The current in the solenoid is reduced. Fig. 6.2 shows the variation of the magnetic flux density  $B$  at right angles to the plane of the coil with time  $t$ .



**Fig. 6.2**

- (i) Use Fig. 6.2 to explain why the induced e.m.f. across the ends of the coil has a constant value from  $t = 0 \text{ s}$  to  $t = 2.5 \text{ s}$ .

.....

.....

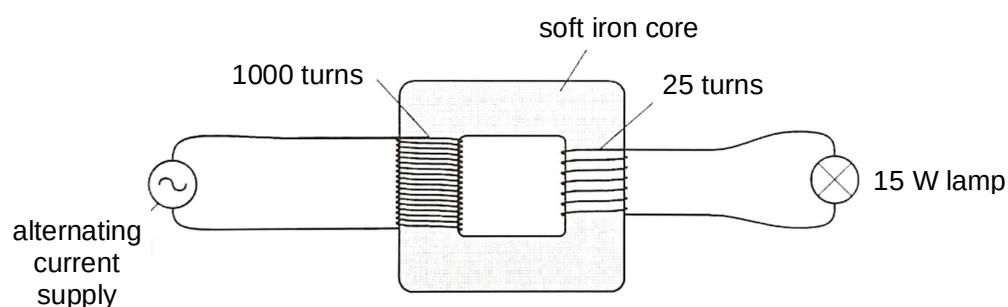
.....

..... [2]

- (ii) Calculate the magnitude of the charge that flows through the coil from  $t = 0$  s to  $t = 2.5$  s.

charge = ..... C [3]

- (b) Fig. 6.3 shows a transformer circuit.



**Fig. 6.3**

The primary coil has 1000 turns and the secondary coil 25 turns. A lamp is connected to the output of the secondary coil. The potential difference across the lamp is 6.0 V and the lamp dissipates 15 W. The transformer has an efficiency of 100%.

- (i) Calculate the current in the primary circuit.

current = ..... A [2]

- (ii) The alternating voltage supply is replaced by a battery.

Explain why the p.d. across the lamp increases but becomes zero some time after the battery is connected.

.....  
 .....  
 ..... [2]

**7 (a)** Some of the energy levels of a particular atom X are shown in Fig. 7.1